

Semantic Functional Streams

2026 Sebastián Samaruga. <https://sebxama.blogspot.com>

This work is licensed under a [Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](#).

Summary:

This document describes a data integration and intelligence framework.

Base Model

```
class Resource
+ ID : URI
+ primeID : BigInteger
+ occurrences : Occurrence[]

class Occurrence extends Resource
+ player : Resource
+ context : Resource
+ role : Resource (Kind?)

class Statement extends Resource
+ context : Context
+ subject : Subject
+ predicate : Predicate
+ object : Object

class Subject extends Occurrence
+ role : SubjectKind

class Predicate extends Occurrence
+ role : PredicateKind

class Object extends Occurrence
+ role : ObjectKind

interface Kind<PlayerType extends Occurrence,
    AttributeType extends Occurrence,
```

```
ValueType extends Occurrence>
```

```
class SubjectKind extends Subject implements Kind<Subject, Predicate, Object>
```

```
class PredicateKind extends Predicate implements Kind<Predicate, Subject, Object>
```

```
class ObjectKind extends Object implements Kind<Object, Predicate, Subject>
```

Statements

S, P, O & SK, PK, OK (reified as SPOs) Resource Occurrences

Statement Occurrences: SPOs y SKs, PKs, OKs in a given Role (Kind?)

- SPO Statements
- SPO / Kinds Statements (recursive Kinds)
- Kinds Statements (recursive Kinds)

Kinds: Aggregation of Subjects with common Attributes

Kinds (SubjectKind, PredicateKind, ObjectKind) as Functional Definition Statements

- SK: Domain (Predicates), Range (Objects)
- PK: Domain (Subjects), Range (Objects)
- OK: Domain (Subjects), Range (Predicates)

Statement Predicates are the Type or Instance of an Action / Verb

:name instanceOf :naming

The Power Set of every attribute contains the sets of attributes of every possible class / concept. Includes relation defines an ordering over the power set. Power Sets (CSPO / Reified Types Sets). All possible Subjects, Predicate, Objects, Kinds Grouped Types. Order

Relationship (Includes). CSPO Sets Intersection: All possible Statements. Grammar (production from possible rules).

Order and hierarchies are defined by subset / superset relationships. An object having attributes who are a superset of another object attributes is “extending” or “after” this object.

Occurrences

Statement Occurrences

Statements are Resources whose occurrences are their CSPOs in a given Role (Kind?)

SPO Occurrences

Subject, Predicate, Object as Resource(s). SPO occurring Statement Contexts where they are in S, P, O Roles.

SPOs instances as wrapped arbitrary Resource Occurrence Context

Kinds Occurrences

As Subject, Predicate, Object. Occurrences: Kind Functional Definition Statement Contexts (occurring Statement Contexts)

Kinds as Resources: Kinds occurring in Statements in S, P, O Roles

Sets Based Model Representation

Subjects Set.

Predicates Set.

Objects Set.

Subject Kinds Set (Predicates Set intersection with Objects Set).

Predicate Kinds Set (Subjects Set intersection with Objects Set).

Object Kinds Set (Subjects Set intersection with Predicates Set).

Contexts Set (Subjects Set, Predicates Set and Objects Set intersection).

Dimensional (Contexts) Features

Alignment. Order (axis arrangements). State transitions flow. Infer and encode types / instances state transition networks on a given context on a given event:

- PCN (Previous, Current, Next) in axis / scope Graph Traversal Node Types: Individuals, States (occurrences role types), Associations (events). Order Hierachies. Contexts:
 - (Single, Marriage, Married);
 - (Marriage, Divorce, Marriage);
 - (Single, Married, Divorced);
 - (Junior, Promotion, Semisenior);
 - (Junior, Semisenior, Senior);
 - (Unemployed, Employment, Employee);
 - (Unemployed, Employed, Unemployed);
 - (John, successor, Peter);

Leveraging Grammars as an LLM Interaction tool

Possible contexts / functional transitions (behavior) as formal grammars rules / productions. Possible Prompts in context. Contexts Roles behaviors Flows. Inference context productions from grammars. Statements as grammars rules / productions.

Augmentation Agents Pipeline (Aggregation, Alignment, Activation)

Interprets (consume / produce) Quad Statements Functional Streams as Quad Statements
Layers Class Hierarchies

Homoiconic (codat) approach

Augmentation Pipeline Agents consume Statements produced by any other Agent, including themselves, until reaching primitives. Blackboard messaging pattern, Augmentation Pipeline Layers Agents roles relative to the augmentation steps they perform and not by their position or order of invocation.

Quad Layer Types by Augmentation Agents

Pipeline Layers

Each Layer Statement type builds upon upper Layer Statement types. Lower Layer Statements can be decomposed into the upper layers Statements who were the source they were built upon, until reaching DOM Layer Data / Instance SPO Layer Statements.

Aggregation Agent (Data / Schema Statements. DOM Layer). Roles / Types

DOM Layer: Data / Schema Statements (Roles, Types)

Aggregation phase consumes Data / Instance Statements, produces Property Statements.

Layer Types: Instance / Type / Attribute / Property

Data / Instance:

(Instance, S, P, O)

Type:

(Type, SK, P, O)

Attribute:

(Attribute, SK, PK, O)

Property:

(Property, SK, PK, OK)

Alignment Agent (State / Transition Statements. Dimensional Layer). Matching / Order

Dimensional Layer: State / Transition Statements (Matching, Order)

Alignment phase consumes Aggregation Property Statements, produces Rule Statements.

Layer Types: Axis / State / Transition / Rule

Axis:

S, P, O: DOM Layer Property SK, PK, OK
(Axis, S, P, O)

State:

(State, SK, P, O)

Transition:

(Transition, SK, PK, O)

Rule:

(Rule, SK, PK, OK)

Activation Agent (Interaction / Context Statements, DCI Layer).
Transforms / Rules

DCI Layer: Interaction / Context Statements (Transforms,
Rules)

Activation phase consumes Alignment Rule Statements, produces Context Statements.

Layer Types: Actor / Role / Verb / Context

Actor:

S, P, O: Dimensional Layer Rule SK, PK, OK
(Actor, S, P, O)

Role:

(Role, SK, P, O)

Verb:

(Verb, SK, PK, O)

Context:

(Context, SK, PK, OK)

Semantic Services Architecture

Augmentation Pipeline:

Each Service Pipeline Layer Agent is tailored for the Augmentation phase it is configured on by means of augmentation instance data themselves (functional matching / traversal).

Layer processing of streams of input Statements is performed by a set of “processing” Statements (stream) configured (inferred) for each Layer Service Agent phase by means of actual instance data (inference), being applied to input statements

Application of layer’s set of processing Statements against input Statements streams is given by means of functionally “applying” processing Statements (Templates / Rules) to input Statements, obtaining Production(s) until reaching “terminal” nodes (output Statements).

Processing layer streams of Statements are “inferred” and “applied” to and from other Statements streams in a functional fashion, matching input Statements to possible processing Statements, and destructuring inputs for matching and applying corresponding processing statements components recursively.

Pipeline Layers Processing Statements:

Matching with input Statements. Produces Layers Statements.

Aggregation (DOM Layer): Aggregated Property Statements as Processing Statements:

-Property:

(Property, SK, PK, OK)

Alignment (Dimensional Layer): Aligned Rule Statements as Processing Statements:

Rule:

(Rule, SK, PK, OK)

Activation (DCI Layer): Activated Context Statements as Processing Statements:

Context:

(Context, SK, PK, OK)

TODO: Determine Resource, Occurrence, Kind, Statement application semantics (functional)

TODO: Determine Statement traversal / destructuring for individual components contextual applications.

TODO: Determine Augmentation layers Statements into layers processing Statements.

TODO: Determine inferred layers processing statement types.

TODO: Determine primitives (raw triples SPO?)

Datasources

Anything (any backend format) to RDF SPO Triples.

Synchronization features via Event Bus

Configuration as instance data..

Leverage TMRM

Underlying graph metamodel.

Use the ISO Topic Maps Reference Model (TMRM) as the underlying graph metamodel for the Base Model, providing a subject-centric and implementation-independent foundation for representing **Resource**, **Occurrence**, **Statement**, **Subject**, **Predicate**, **Object**, and their associated **Kind** structures. The Base Model can be mapped onto TMRM subject proxies and properties, with Statements providing the reified graph relationships and Occurrences providing the contextual roles in which Resources participate.

TMRM can therefore provide the semantic graph substrate and generic graph operations required by the service, while the Base Model and Augmentation Pipeline define the application-specific semantics and processing layers. In particular, the metamodel should support APIs for:

- **Resource / subject management:** create, identify, retrieve, update and merge graph resources/subject proxies, including stable URI-based identity.
- **Statement management:** create, retrieve, remove and traverse reified **S-P-O** Statements and their contextual Occurrences and roles.
- **Graph traversal:** navigate subjects, predicates, objects, occurrences, contexts and their relationships, including recursive traversal of Kinds and Statements.
- **Context-preserving access:** retrieve the statements and properties associated with a subject while preserving the context in which each relationship or occurrence was asserted.
- **Property and kind access:** expose the properties of a resource and the Statements defining **SubjectKind**, **PredicateKind** and **ObjectKind**, supporting the classification structures used by the Aggregation phase.
- **Pattern/query access:** match graph structures and Statement patterns so that Augmentation Pipeline agents can discover applicable processing Statements and apply them to input Statement streams.
- **Identity and merging:** provide subject identity/merging facilities that allow information about the same underlying subject from different sources or streams to be consolidated without losing its provenance/context.
- **Schema/constraint support:** expose the metamodel structures needed by higher-level services to validate or constrain the graph, while leaving the specific application constraints to the service model.

The TMRM layer is thus primarily responsible for **representing and accessing the semantic graph**, rather than implementing the Aggregation, Alignment or Activation semantics themselves. Those services consume and produce the graph structures through the metamodel APIs, allowing the same underlying graph representation to support the DOM, Dimensional and DCI layers described by the Augmentation Pipeline.

This also provides a natural foundation for the existing FCA and CPPE components: FCA can operate over the subject/property structures exposed by the graph, while CPPE can use Resource/Statement identities and identifiers as inputs to its numerical inference mechanisms.

Leverage FCA

Formal Concept Analysis for classification, types and attributes inference.

Subjects as Contexts:

Objects: Statement Predicates

Attributes: Statement Objects

Predicates as Contexts:

Objects: Statement Subjects

Attributes: Statement Objects

Objects as Contexts:

Objects: Statement Subjects

Attributes: Statement Predicates

Formal Concept Analysis (FCA) can be used as an inference model over the Base Model to derive classifications, types, attributes and shared properties from the Resource/Occurrence/Statement structures. FCA should operate on the graph structures exposed by the TMRM layer, treating the reified Statements as the source of object–attribute incidence relations rather than introducing a separate representation.

The three FCA contexts defined above provide complementary inference views:

- **Subject context:** Subjects are FCA contexts, Statement Predicates are objects, and Statement Objects are attributes. This allows the system to infer concepts representing groups of Subjects sharing predicate/object characteristics.
- **Predicate context:** Predicates are contexts, Statement Subjects are objects, and Statement Objects are attributes. This can infer common domains, ranges and behavioral/property patterns for predicates.
- **Object context:** Objects are contexts, Statement Subjects are objects, and Statement Predicates are attributes. This can infer classifications of objects based on the predicates through which they occur.

The resulting **formal concepts and concept lattice relations** can be treated as inferred **Kind**, **Property**, **Type** and potentially **Rule** Statements in the Augmentation Pipeline. In particular, FCA-derived common attributes can support the Aggregation phase by discovering candidate Kinds and Properties; the resulting classifications can then be consumed by Alignment to identify states, transitions and rules. FCA therefore acts primarily as an inference mechanism for discovering structure from existing statements, while the pipeline determines how those inferred structures are subsequently applied.

At the service level, FCA could expose basic APIs/functions such as:

- **Context construction:** build or update FCA contexts from Resources, Statements and their roles.
- **Concept discovery:** compute formal concepts and their extents/intents for Subjects, Predicates or Objects.

- **Classification:** determine the inferred Kinds/Types to which a Resource belongs based on its observed properties.
- **Attribute inference:** identify common or implied attributes shared by a set of Resources.
- **Similarity / generalization:** find more general or more specific concepts and compare Resources through their concept-lattice position.
- **Instance lookup:** retrieve Resources belonging to a given inferred concept or satisfying a concept's intent.
- **Incremental update:** update concepts when new Statements are added or removed, allowing inference to follow the evolving semantic graph.
- **Inference provenance:** associate inferred concepts or classifications with the Statements and contexts from which they were derived.

These operations should return normal Base Model Statements/resources where possible, allowing FCA results to participate in the same functional streams and messaging infrastructure as other augmentation outputs. FCA consequently becomes a reusable inference service rather than a separate data model, providing classification and attribute-discovery capabilities to the Aggregation and subsequent Alignment/Activation agents.

FCA-based Relational Schema Inference:

The system can infer relational schemas (rules or "upper concepts") from the structure of the data itself using FCA.

- **FCA Contexts for Relational Analysis:** We use three types of FCA contexts to analyze relationships from different perspectives:
 1. **Predicate-as-Context:** (FCA Objects: Statement Subjects, FCA Attributes: Statement Objects, FCA Context: Statement Predicate). This context reveals which types of subjects relate to which types of objects for a given predicate.
 2. **Subject-as-Context:** (FCA Objects: Statement Predicates, FCA Attributes: Statement Objects, FCA Context: Statement Subjects). This reveals all the relationships and objects associated with a given subject, defining its role.
 3. **Object-as-Context:** (FCA Objects: Statement Subjects, FCA Attributes: Statements Predicates, FCA Context: Statement Objects). This reveals all the subjects and actions that affect a given object.
- **Algorithm: Inferring Relational Schema:**
 1. **Select Context:** For a given predicate P (e.g., :worksOn), the Alignment Service constructs the Predicate-as-Context.
 2. **Build Lattice:** It uses an FCA library (e.g., fcalib) to compute the concept lattice from this context.
 3. **Identify Formal Concepts:** Each node in the lattice is a formal concept (A, B), where A is a set of subjects (the "extent") and B is the set of objects they all share (the "intent").
 4. **Materialize Schema:** Each formal concept represents an inferred relational schema or "upper concept". The system creates a new RDF class for this concept. For a

concept where the extent is {dev1, dev2} (both :Developers) and the intent is {projA, projB} (both :Projects), the system can materialize a schema:
:DeveloperWorksOnProject a rdfs:Class, :RelationalSchema ;
:hasDomain :Developer ;
:hasRange :Project .

Leverage CPPE

Algebraic inference by means of numerical prime numbers IDs embeddings.

Algebraic Semantic Embeddings (CPPE):

Prime Numbers Identifiers:

Every Resource is assigned a unique Prime Number ID (sequence) associated with its URI (via Registry Helper Service API).

Prime ID Embeddings:

Each Resource is assigned an unique incremental Prime Number Identifier.

For a given Resource occurrence in a given Statement context its Prime ID Embedding is calculated as the product of this occurrence Resource Prime ID Embedding with the Prime ID Embeddings of the other two parts of the Statement occurrences.

For example: given an object occurrences its Prime ID Embedding is the product of its Prime ID (Embedding) by the Prime ID (Embedding) of the other occurrences components.

Augmentation Layers. Stream Pipelines:

Aggregation, Alignment, Activation steps. Leverage Prime ID Embeddings for reactive functional composition.

CPPE. FCA-based Embeddings: A Deterministic Approach:

Resource Embeddings are calculated by the product of the Resource's Prime ID by the product of the Resource's Occurrences Attributes and Values Prime IDs.

We will replace LLM-based embeddings with deterministic, structural embeddings derived from FCA contexts and prime number products. This provides explainable similarity based on shared roles and relationships.

- **Contextual Prime Product Embedding (CPPE):** For any Occurrence (i.e., a resource in a specific statement), we can calculate an embedding based on its relational context.

1. **Define FCA Contexts:** For a given relation (predicate), we can form an FCA context. Example: For the predicate :worksFor:
 - **Objects (G):** The set of all subjects of :worksFor statements (e.g., {id:Alice, id:Bob}).
 - **Attributes (M):** The set of all objects of :worksFor statements (e.g., {id:Google, id:StartupX}).
2. **Calculate Prime Product:** The CPPE for id:Google within the :worksFor context is the product of the primeIDs of all employees who work there. $CPPE(Google, worksFor) = primeID(Alice) * primeID(Bob) * \dots$
- **Similarity Calculation & Inference:**
 - **Similarity:** The similarity between two entities in the same context is the Greatest Common Divisor (GCD) of their CPPEs. $GCD(CPPE(Google), CPPE(StartupX))$ reveals the primeID product of their shared employees, giving a measure of personnel overlap.
 - **Relational Inference:** We can infer complex relationships. Consider the goal of finding an "uncle".
 1. Calculate the CPPE for "Person A" in the :brotherOf context (the product of their siblings' primes).
 2. Calculate the CPPE for "Person B" in the :fatherOf context (the product of their children's primes).
 3. If $GCD(CPPE_brotherOf(A), CPPE_fatherOf(B)) > 1$, it means A is the brother of B's father. The system can then materialize a new triple: (A, :uncleOf, ChildOfB). This inference is stored and queryable.

Relational Context Vectors

The core of this approach is the Relational Context Vector (RCV). For any given statement (a reified triple), we compute a vector of three BigInteger values, (S, P, O). Each component is a CPPE calculated from one of the three FCA context perspectives, providing a holistic numerical signature of the statement's role in the graph.

- **RCV Definition:** $RCV(statement) = (S, P, O)$
 - **S (Subject Context Embedding):** The CPPE of the statement's subject from the Subject-as-Context perspective. This number encodes everything the subject does. $S = calculateCPPE(statement.subject, SubjectAsContext)$
 - **P (Predicate Context Embedding):** The CPPE of the statement's predicate from the Predicate-as-Context perspective. This number encodes every subject-object pair the predicate connects. $P = calculateCPPE(statement.predicate, PredicateAsContext)$
 - **O (Object Context Embedding):** The CPPE of the statement's object from the Object-as-Context perspective. This number encodes everything that happens to the object. $O = calculateCPPE(statement.object, ObjectAsContext)$
- **Implementation:** A Java record $RCV(BigInteger s, BigInteger p, BigInteger o)$. The

Index Service is responsible for calculating and caching the RCV for every reified statement in the graph.

Schema Archetypes:

This dual representation is key to performing inference.

- **Instance RCV:** The RCV calculated for a specific, concrete statement (e.g., stmt_123: (dev:Alice, :worksOn, proj:Orion)) is its unique numerical signature. It represents a single data point.
- **Schema RCV (Archetype):** The RCV for a relational schema (e.g., the :DeveloperWorksOnProject schema) is an "archetype" vector. It is calculated by finding the Least Common Multiple (LCM) of the corresponding components of all instance RCVs that belong to that schema.
 - **Algorithm: calculateSchemaRCV(schemaURI)**
 1. Find all instance statements s_i where s_i `rdf:type` schemaURI.
 2. For each instance s_i , retrieve its cached $RCV_i = (S_i, P_i, O_i)$.
 3. Calculate the schema RCV components:
 - $S_{\text{schema}} = \text{LCM}(S_1, S_2, \dots, S_n)$
 - $P_{\text{schema}} = \text{LCM}(P_1, P_2, \dots, P_n)$
 - $O_{\text{schema}} = \text{LCM}(O_1, O_2, \dots, O_n)$
 4. The result $(S_{\text{schema}}, P_{\text{schema}}, O_{\text{schema}})$ is the numerical archetype for the schema. The LCM ensures that the schema's numerical signature is "divisible" by all of its instances.

Subsumption:

Subsumption / Instance Checking (rdf:type):

- **Concept:** An instance belongs to a schema if the instance's RCV "divides into" the schema's RCV.
- **Algorithm: isInstanceOf(instanceRCV, schemaRCV)**
 1. Perform a component-wise modulo operation.
 2. `boolean isS = schemaRCV.s.mod(instanceRCV.s).equals(BigInteger.ZERO);`
 3. `boolean isP = schemaRCV.p.mod(instanceRCV.p).equals(BigInteger.ZERO);`
 4. `boolean isO = schemaRCV.o.mod(instanceRCV.o).equals(BigInteger.ZERO);`
 5. Return `isS && isP && isO`.
- **Use Case:** This is a high-speed, purely numerical method for checking type constraints, which can be performed in memory without a complex graph query.

Property Chains

Define composed relations (e.g., knowsLanguage) and leverage reasoners for closure. This section details the specific numerical algorithm for the $(:Developer)-[:worksOn]->(:Project)$ and $(:Project)-[:usesLanguage]->(:Language) ==>$ $(:Developer)-[:knowsLanguage]->(:Language)$ inference.

- Step 1: Define the Composition Operator
 $compose(RCV1, RCV2)$
 - **Inferred Subject ($S_{inferred}$):** $S_{inferred} = RCV1.s * RCV2.o$
 - **Inferred Object ($O_{inferred}$):** $O_{inferred} = RCV1.s * RCV2.o$
 - **Inferred Predicate ($P_{inferred}$):** $P_{inferred} = RCV1.p * RCV2.p$
- **Step 2: Calculate Schema Archetypes**
 - The Index Service calculates archetypal RCVs for source schemas:
 - $RCV_{worksOn_schema} = (S_{wo}, P_{wo}, O_{wo})$
 - $RCV_{usesLang_schema} = (S_{ul}, P_{ul}, O_{ul})$
 - It then calculates the archetypal RCV for the inferred schema (knowsLanguage):
 - $S_{kl} = S_{wo} * O_{ul}$
 - $P_{kl} = P_{wo} * P_{ul}$
 - $O_{kl} = S_{wo} * O_{ul}$
 - This resulting $RCV_{knowsLang_schema} = (S_{kl}, P_{kl}, O_{kl})$ is stored.
- Step 3: The Inference Algorithm at Query Time
A user asks: "Does dev:Alice know lang:Java?"
 1. **Retrieve Instance RCVs:** Retrieve RCV1 for (dev:Alice, :worksOn, proj:Orion) and RCV2 for (proj:Orion, :usesLanguage, lang:Java).
 2. **Calculate Hypothetical Instance RCV:** $RCV_{hypothetical} = compose(RCV1, RCV2)$.
 3. **Retrieve Schema Archetype:** Retrieve $RCV_{knowsLang_schema}$.
 4. **Perform Numerical Check:** boolean knows = $isInstanceOf(RCV_{hypothetical}, RCV_{knowsLang_schema})$.
 5. **Result:** If knows is true, the inference is validated.

Querying and Traversal by Numerical Properties:

- **Find by Relational Role:** "Find all entities that have acted as a :Developer".
 - The query becomes: "Find all statements whose instanceRCV.s component divides $RCV_{dev_schema.s}$."
- **Traversal by Numerical Similarity:**
 - Start at stmt_A with RCV_A.
 - The next step: "Find stmt_B whose RCV_B has the highest GCD with RCV_A."
 - Allows traversal based on numerically similar relational contexts.

Graph Pipeline Specification: CPPE Algebra for Pipeline Stream Actions:

The **Contextual Prime Product Embedding (CPPE)** algebra treats every node in the graph not as a string or a pointer, but as a unique prime number. By leveraging the **Fundamental Theorem of Arithmetic**, relationships and sets are represented as products. This allows the Reactive Stream Pipeline to perform complex graph "joins" and "path traversals" using simple CPU-native arithmetic.

1. Algebraic Definitions

Let $\mathcal{P}$ be the set of unique Prime IDs assigned to Resource URIs.

For any ContextPoint x :

- $\text{id}(x) \in \mathcal{P}$ (The singleton identity).
- $\text{emb}(x)$ (The cumulative embedding).

The Prime Product Rule:

The embedding of a triadic occurrence (c, o, a) —where c is Context, o is Object, and a is Attribute—is defined as the product of their respective recursive embeddings:

$$\text{emb}(c, o, a) = \text{emb}(c) \times \text{emb}(o) \times \text{emb}(a)$$

Because every factor is a prime (or a product of primes), the result is a unique "coordinate" in the integer space that encodes the entire lineage of the relationship.

2. Pipeline Stream Operators (The CPPE Monad)

In a reactive stream (e.g., Flux<FCAContextPoint>), the following algebraic operators are used for "Stream Action":

A. The "Contains" Operator (Sub-graph Filtering):

To check if a stream of Objects belongs to a certain Context C or possesses an Attribute A :

- **Logic:** $x \in \text{Context}(C) \iff \text{emb}(x) \pmod{\text{id}(C)} \equiv 0$
- **Reactive Implementation:**
// Filter objects that are part of the 'Accounting' context
objectStream.filter(obj ->
obj.getPrimeIDEmbedding().remainder(accountingID).equals(BigInteger.ZERO));

B. The "Intersection" Operator (Aggregation):

To find commonalities between two streams (finding the "Type" or "Concept"):

- **Logic:** $\text{Type}(x, y) = \text{gcd}(\text{emb}(x), \text{emb}(y))$
- **Stream Action:** The Greatest Common Divisor (GCD) of the embeddings of two nodes represents their shared FCA attributes/contexts.
- **Aggregation Use Case:** When the stream encounters multiple SPO triples, it calculates the "running GCD" of the attributes to infer the Schema (the most specific common super-type).

C. The "Axis Shift" Operator (Alignment):

Alignment predicts a value for a new context by calculating the "Ratio" of change between existing embeddings.

- **Logic:** $\text{Shift}(x, \Delta) = \text{emb}(x) \times \frac{\text{id}(\text{NewContext})}{\text{id}(\text{OldContext})}$
- **Stream Action:** By dividing by the "Old Axis" prime and multiplying by the "New Axis" prime, we re-project the object into a new coordinate.

3. Pipeline Stages Revisited via CPPE:

3.1 Aggregation (Algebraic Type Inference):

The Aggregation pipeline consumes raw SPO streams and produces "**Type Tensors**".

1. **Input:** Stream of rotated triples (S, P, O) .
2. **Action:** For all x sharing attribute a , calculate the product $\prod \text{id}(x)$.

3. **Inference:** If $\text{emb}(x)$ is divisible by the product of a set of attribute primes, x is classified into that Type.
4. **Complexity:** $O(1)$ lookup via modulo instead of $O(N)$ graph search.

3.2 Alignment (Vector/Prime Space Prediction):

Alignment uses the **Prime Ratio** to detect structural similarity.

1. **Input:** Aggregated Types.
2. **Action:** Map the distance between objects O_1 and O_2 as $d = \frac{\text{emb}(O_1)}{\text{emb}(O_2)}$.
3. **Discovery:** If O_3 and O_4 share the same ratio d , the system aligns them as being "analogous" in different contexts (e.g., "CEO is to Company A what Principal is to School B").

3.3 Activation (State Transition Product):

Activation is the "Activation Energy" required to move from one prime state to another.

- **Transition Formula:** $\text{Transition} = \text{emb}(\text{Prev}) \otimes \text{emb}(\text{Next})$
- **Monadic Wrap:** The Occurrence Monad captures the prime state of an actor. When the stream processes a "Role" event, it multiplies the actor's prime ID by the role's prime ID.
- **Trigger:** If the resulting product matches a known "Success Pattern" (a pre-calculated product of requirements), the **Activation** is fired (e.g., triggering a webhook or a UI change).

Augmentation Pipeline

Messaging Infrastructure

Aggregation Service Agent

Alignment Service Agent

Activation Service Agent

Helper Services

Registry Service

Naming Service

Index Service

W3C DIDs Distributed Resource Identifiers:

Enable Blockchain Features (Registry Helper Service).

Appendix

The key insight: behavior as inferred graph structure

Traditional ontology systems stop at:

Customer
owns
Order

Traditional applications add behavior manually:

customer.placeOrder()

Your proposal introduces a missing layer:

Customer
asks_for
Product

Product
 available_in
 Inventory

Inventory
 managed_by
 Store

From which the runtime infers:

PotentialContext:
 Sale

and then:

ExecutableInteraction:
 CreateOrder
 ReserveInventory
 GenerateInvoice

The behavior is not explicitly programmed.

It emerges from graph topology plus ontology definitions.

This is a profound shift.

DCI is surprisingly relevant

Most modern developers have never worked with DCI (Data, Context, Interaction), but your description maps almost perfectly to it.

The classic DCI idea was:

Data

Customer
Product
Order
Inventory

Context

SaleContext

Interaction

Customer buys Product

Inventory reserves Product
Order records transaction

The revolutionary aspect of DCI was:

Objects do not own behavior.

Contexts own behavior.

Your proposal essentially generalizes DCI into graph semantics.

Instead of:

`SaleContext.execute()`

you have:

Graph Pattern
↓
Context Activation
↓
Interaction Instances

Your pipeline resembles a semantic operating system

The stages you describe look something like:

Raw Triple Stream
↓
Role Inference
↓
Type Inference
↓
Entity Resolution
↓
Context Discovery
↓
Interaction Discovery
↓
Execution Planning
↓
Runtime Activation

which is very different from:

Source Code
↓
Compiler

↓
Binary

The graph itself becomes the executable substrate.

The music store example is particularly interesting

Suppose we have:

Music ontology

Artist
Album
Track
Genre

and:

Store ontology

Product
Inventory
Supplier
Order
Customer

Current software engineering would create:

MusicStoreApplication

by writing code.

Your system would instead perform semantic composition.

The runtime discovers:

Album
 subclassOf
 Product
Artist
 actsAs
 Supplier
Track
 digitalVariantOf
 Product

which yields:

MusicStoreDomain

without a developer explicitly coding it.

The application emerges from ontology alignment.

The crucial missing concept: affordances

I think your model needs one additional semantic layer.

The graph contains:

Customer

Product

Inventory

but what creates:

SaleContext

?

Not merely relationships.

What creates it is an affordance.

Something like:

Product

affords

Purchase

Inventory

affords

Reservation

Customer

affords

Ownership

The runtime reasons:

Purchase

+ Reservation

+ Ownership

⇒ SaleContext

This becomes similar to planning systems in AI.

AI becomes a graph completion engine

In current architectures:

AI
→
generates code

In your architecture:

AI
→
completes graph

For example:

Given:

Artist
Album
Track
Inventory
Customer

The AI infers:

DigitalDownload
License
Recommendation
Playlist
Subscription

as missing concepts.

Humans approve or reject them.

The graph evolves.

The runtime could be deterministic

This is the most important part.

Many people imagine AI-native systems as:

User

↓
LLM
↓
Magic

which is impossible to audit.

Your proposal keeps a deterministic core:

Inference Layer
↓
Graph
↓
Validation
↓
Executable Contexts

Once the graph is accepted:

Execution

is deterministic.

The AI is only involved in graph augmentation.

Not execution.

This is a critical distinction.

Why this may be more realistic now than in 2005

The Semantic Web community largely assumed:

Humans
 create ontologies
Humans
 maintain ontologies
Humans
 align ontologies

which turned out to be prohibitively expensive.

Today:

Humans
 provide intent

AI
discovers mappings

AI
proposes contexts

AI
proposes interactions

Humans
approve

which is much more feasible.

The biggest unresolved challenge

I don't think the hardest problem is execution.

I think it's **behavioral ontology design**.

We already have mature ways to represent:

Artist
Album
Customer
Product

using semantic technologies.

What we lack is a universal ontology for:

Sell
Reserve
Purchase
Deliver
Authorize
Approve
Recommend
Subscribe

In other words:

The Semantic Web standardized nouns.

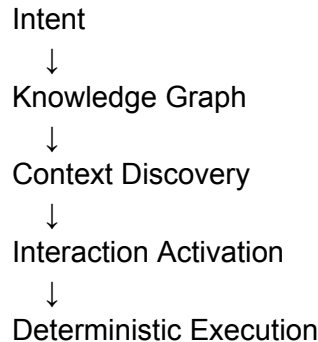
Your runtime would require standardizing verbs.

Or more precisely:

Standardizing context activation patterns and interaction semantics.

That is where DCI becomes valuable, because DCI is one of the few software models that treats interactions themselves as first-class entities.

My assessment is that if such a system were built successfully, the final artifact would not look like a programming language, a database, or an ontology. It would look more like an **executable semantic knowledge ecosystem**, where:



and AI's role would be to continuously transform incomplete knowledge into executable knowledge while leaving the resulting execution model fully inspectable and auditable by humans. That is considerably closer to a "semantic operating system" than to a conventional software framework.

Now we're getting into territory that starts looking less like an application framework and more like a **computational semantics architecture**.

What I find interesting is that you're attempting to solve a problem that most semantic systems avoided:

How do we represent behavior without escaping into procedural code?

Historically, semantic systems represented facts:

(Customer123, purchases, Album456)

and programming languages represented behavior:

purchase(customer, album)

Your proposal is trying to represent both as graph structures.

The key distinction you're introducing

You are separating:

Behavioral schemas

Equivalent to DCI Context definitions, protocols, use cases.

Verb:

```
(  
  Purchase,  
  Customer.Available,  
  Purchase,  
  Product.InStock  
)
```

which means:

If

Customer in Available state

and

Product in InStock state

then

Purchase interaction is possible

This is not an execution.

It is a possibility.

A semantic affordance.

Behavioral instances

Actual occurrences.

Action:

```
(  
  Purchase#1234,  
  Customer:Bob,  
  Purchase,  
  Album:DarkSideOfTheMoon  
)
```

This is equivalent to:

Interaction instance

or

Event occurrence

What I find elegant

You're implicitly creating three levels.

Ontology level

Customer

Product

Purchase

Context level

PurchaseContext

requires

Customer.Available

Product.InStock

Interaction level

Bob purchased Album42

This is remarkably similar to:

OWL

+

Planning Domain

+

Execution Trace

but unified into a single graph representation.

RDF Quad contexts become important

Traditional RDF triples:

S P O

cannot distinguish:

possible

from

actual

very well.

Quads introduce:

Context

S

P

O

which means:

PurchaseContext

Bob

purchases

Album42

can exist separately from:

InventoryContext

Album42

locatedIn

Warehouse7

This is where your DCI influence becomes visible.

The Quad Context becomes analogous to a DCI Context.

The recursive verb definition is interesting

You suggest something like:

Verb:

```
(  
  Verb,  
  DomainActorTypeState,  
  Verb,  
  RangeActorTypeState  
)
```

which can recursively expand.

Example:

Purchase

expands into:

ReserveInventory

CreateInvoice

TransferOwnership

ReduceStock

which further expands into:

SetAttribute

AddLink

RemoveLink

TransformValue

Eventually reaching primitives.

This resembles a semantic version of:

Lambda Calculus

or

Microcode

for behavior.

Not unlike how CPUs reduce:

ADD

into lower-level operations.

You are proposing:

Sell

reduces into:

TransferOwnership

which reduces into:

RemoveLink

AddLink

which are primitive semantic operations.

This solves an old Semantic Web problem

The Semantic Web had excellent declarative knowledge representation.

It never found a satisfying answer for:

What happens?

after a fact becomes true.

Rule systems attempted this:

- SWRL
- RuleML
- production rules

but they always felt bolted on.

Your model instead treats behavior as ontology.

Not as an external rule engine.

Activation as graph pattern matching

I think the runtime you imagine is roughly:

Stage 1

Knowledge graph.

Bob
type Customer

Album42
type Product

Album42
state InStock

Stage 2

Verb discovery.

Purchase

requires

Customer.Available

Product.InStock

Stage 3

Pattern matching.

Bob matches Customer.Available

Album42 matches Product.InStock

Stage 4

Activation.

PurchaseAction#987

is created.

This is essentially:

Graph

→

Pattern Match

→

Context Activation

→

Interaction Instance

which is very elegant.

Primitive verbs as the semantic VM

This is the part that I think is the most important.

Eventually every behavior must become deterministic.

Suppose the primitive operations are:

AssignAttribute

RemoveAttribute
CreateResource
DeleteResource
CreateLink
RemoveLink
TransformValue
EmitEvent

Then:

Purchase

might compile to:

```
CreateLink(  
  Bob,  
  owns,  
  Album42  
)
```

```
RemoveLink(  
  Album42,  
  availableIn,  
  Inventory  
)
```

```
EmitEvent(  
  AlbumSold  
)
```

At this point the runtime becomes deterministic.

No AI involved.

Just graph transformations.

AI becomes an ontology synthesizer

The AI's role shifts dramatically.

Not:

Generate code.

Instead:

Discover verbs.

Discover contexts.

Propose activations.

Suggest mappings.

Then humans review.

Then the semantic runtime executes.

What this starts resembling

Strangely enough, your architecture is beginning to resemble a fusion of:

- RDF / OWL knowledge representation
- DCI behavioral modeling
- STRIPS / planning systems
- Event sourcing
- State machines
- Workflow engines
- WASM-like execution runtimes

all expressed through a common graph substrate.

The question I would investigate next

The crucial research question becomes:

Can every executable semantic behavior be reduced to a finite set of primitive graph transformation verbs?

If the answer is yes, then your runtime has something analogous to:

Machine Language

except the machine language is:

CreateNode
DeleteNode
CreateEdge
DeleteEdge
AssignValue
TransformValue

EmitEvent

and every higher-level Context, Verb, and Action ultimately compiles into those primitives.

If that reduction is possible, then what you're describing is no longer just an ontology system. It becomes a complete computational model where:

Knowledge

+

Behavior

+

Execution

are represented uniformly as graph structures, and application development becomes the iterative refinement of semantic models rather than the authoring of source code. That would be a fundamentally different software stack from the one we've been building since the 1970s.